

**MT-2005: Probability &
Statistics BSCS (Part A)**
(A, B, C, D, E, F, G, H, J, K)

Tuesday, 28th May, 2024

Course Instructors

Dr. Muhammad Usman Ashraf, Dr. Neelam,
Ms. Amara, Ms. Kiran

Serial No:

Final Exam

Max. Time: 100 min

Total Marks: 60

Signature of Invigilator

Student Name

Roll No.

Section

Signature

Do not write anything on the question paper except the information required above.

Instructions:

1. Read the question carefully, understand the question, and then attempt your answers in the provided answer booklet.
2. Verify that you have **two (2)** printed page of the question paper including this page. There are **six (6)** questions.
3. Calculator sharing is strictly prohibited.
4. **One bonus mark** is given to those students who will solve **each question on separate page**.

Question #1

[10 marks]

There are three identical cards except that both the sides of the first card are coloured red, both sides of the second card are coloured blue and for the third card one side is coloured red and the other side is blue. One card is randomly selected among these three cards and put down, visible side of the card is red. What is the probability that the other side is blue?

Question #2

[12 marks]

Alex takes a quiz in his Anthropology 100 class. The quiz consists of 10 questions, the first 4 are True-False, the last 6 are multiple choice questions with 4 possible answers each, only one of which is correct. Alex did not study for the quiz, so he guesses independently on each question. Find the probability that he answers exactly 2 questions correctly.

Question #3

[04*3=12 marks]

In a casino game chuck-a-luck, three unbiased six-sided dice are rolled. One possible bet is \$1 on fives, and the payoff is equal to \$1 for each five on that roll. In addition, the dollar bet is returned if at least one five is rolled. The dollar that was bet is lost only if no fives are rolled. Let X denote the payoff amount for this game.

a) Determine the probability density function $f(x)$.

b) Calculate mean and variance.

Question #4**[04+04+02=10 marks]**

An advertising company has 12 men and 8 women. Suppose the company needs to select a team of 5 members to work on a commercial for the new hybrid car. If the members of the team are selected at random,

- what is the probability that 3 men and 2 women will be selected?
- What is the probability that men will constitute a majority in the team?
- Find the Variance of a Hypergeometric Distribution such that the probability that a 3-trial hypergeometric experiment results in exactly 2 successes, when the population consists of 7 items.

Question #5**[06 marks]**

Suppose it has been observed that, on average, 180 cars per hour pass a specified point on a particular road in the morning rush hour. Due to impending roadworks it is estimated that congestion will occur closer to the city center if more than 5 cars pass the point in any one minute. What is the probability of congestion occurring?

Question #6**[10 marks]**

A card is lost from a pack of 52 cards. From the remaining cards two are drawn randomly and found to be both clubs. Find the probability that the lost card is also a club?

Question :-

$X \rightarrow$ denote the payoff for this game

$X \rightarrow -1, 1, 2, 3$

$$(a) \quad f(-1) = P(\text{no five}) = {}^3C_0 \left(\frac{1}{6}\right)^0 \left(\frac{5}{6}\right)^3 = 125/216$$

$$f(1) = P(\text{one five}) = {}^3C_1 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = 75/216$$

$$f(2) = P(\text{two five}) = {}^3C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = 15/216$$

$$f(3) = P(\text{three five}) = {}^3C_3 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^0 = 1/216$$

(b)

x	$f(x)$	$x f(x)$	$x^2 f(x)$
-1	$125/216$	$-125/216$	$125/216$
1	$75/216$	$75/216$	$75/216$
2	$15/216$	$30/216$	$60/216$
3	$1/216$	$3/216$	$9/216$
		$-17/216$	$269/216$

$$\mu = E(X) = \sum x_i f(x_i) = -17/216 = -0.07870$$

$$\sigma^2 = \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \frac{269}{216} - \left(\frac{-17}{216}\right)^2 = 1.239176$$

Question

Let E_1 = Lost card is a club

E_2 = Lost card is not a club

A = both drawn are club

$$P(E_1) = \frac{13}{52}, \quad P(E_2) = \frac{39}{52} = \frac{3}{4}$$

$$P(A/E_1) = P(\text{drawing both cards club given that lost card is club}) = \frac{12}{51} \times \frac{11}{50}$$

$$P(A/E_2) = P(\text{drawing both card club given that lost card is not club}) = \frac{13}{51} \times \frac{12}{50}$$

Now

$$\begin{aligned} P(E_1/A) &= \frac{P(E_1 \cap A)}{P(A)} = \frac{P(E_1) P(A/E_1)}{P(E_1) P(A/E_1) + P(E_2) P(A/E_2)} \\ &= \frac{\frac{12}{51} \times \frac{11}{50} \times \frac{1}{4}}{\frac{12}{51} \times \frac{11}{50} \times \frac{1}{4} + \frac{3}{4} \times \frac{13}{51} \times \frac{12}{50}} = \frac{11}{50} \end{aligned}$$

Question

Let RR: card with both side red

BB: card with both side Blue

RB: card with one side red and
other side Blue

A: Event that the visible side of the
chosen card is red

$$P(RR) = \frac{1}{3}, \quad P(BB) = \frac{1}{3}, \quad P(RB) = \frac{1}{3}$$

$$P(A) = P(RR)P(A/RR) + P(BB)P(A/BB) + P(RB)P(A/RB)$$

$$= \frac{1}{3} \times 1 + \frac{1}{3} \times 0 + \frac{1}{3} \times \frac{1}{2} = \frac{1}{3} + \frac{1}{6} \\ = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2}$$

Now

$$P(RB/A) = \frac{P(RB \cap A)}{P(A)} = \frac{P(RB)P(A/RB)}{\frac{1}{2}} \\ = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}$$

Question

(a)

12 Men, 8 women
select = 5

$$P[3 \text{ men, 2 women}] = \frac{{}^{12}C_3 {}^8C_2}{{}^{20}C_5} = 0.3973$$

(b) $P[\text{men will constitute a majority in the team}] = ?$

$$= \frac{{}^{12}C_3 {}^8C_2}{{}^{20}C_5} + \frac{{}^{12}C_4 {}^8C_1}{{}^{20}C_5} + \frac{{}^{12}C_5 {}^8C_0}{{}^{20}C_5} = 0.7038$$

(c) $n \frac{K}{N} \left(1 - \frac{K}{N}\right) \left(\frac{N-n}{N-1}\right) = \text{variance}$
hypergeometric distribution =

$\Rightarrow \begin{matrix} \text{Trials} \\ \downarrow \\ n = 3 \end{matrix}, \begin{matrix} \text{Success} \\ \downarrow \\ K = 2 \end{matrix}, N = 7 \rightarrow \text{Sample Size}$

$$= 3 \times \frac{2}{7} \left(1 - \frac{2}{7}\right) \left(\frac{4}{6}\right) = \frac{3 \times 2}{7} \times \frac{5}{7} \times \frac{4}{6}$$
$$= \frac{20}{49}$$

Question

$X \rightarrow$ number of True False questions answered correctly.

$Y \rightarrow$ number of multiple choice questions answered correctly.

$$q = 1 - p$$

$X \rightarrow$ Binomial distribution $n_1 = 4$, $P_1 = 0.50$, $1 - P_1 = 0.50$

$Y \rightarrow$ " " $n_2 = 6$, $P_2 = 0.25$, $1 - P_2 = 0.75$

$$P(X=x) = {}^n C_x P^x (1-P)^{n-x}$$

$$P(X+Y=2) = P(X=0 \wedge Y=2) + P(X=1 \wedge Y=1) + P(X=2 \wedge Y=0)$$

$$= P(X=0) \times P(Y=2) + P(X=1) \times P(Y=1) + P(X=2) \times P(Y=0)$$

$$= {}^4 C_0 (0.50)^0 (0.50)^4 \times {}^6 C_2 (0.25)^2 (0.75)^4$$

$$+ {}^4 C_1 (0.50)^1 (0.50)^3 \times {}^6 C_1 (0.25)^1 (0.75)^5$$

$$+ {}^4 C_2 (0.50)^2 (0.50)^2 \times {}^6 C_0 (0.25)^0 (0.75)^6$$

$$= 0.17427$$

Question

$$P(X=x) = \frac{e^{-\lambda t} \lambda^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

λ = average number of outcomes per unit time
 t = time interval

$$e = 2.71832$$

$$\mu = \lambda t, \quad P(X=x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, 3, \dots$$

(more than 5 cars in any one min)

$$\lambda = 180 \text{ cars per hour}$$

$$\lambda = 3 \text{ cars per min}$$

$$t = 1 \text{ min}$$

$$P(X > 5) = 1 - \sum_{x=0}^5 \frac{e^{-3} (3)^x}{x!}$$

$$= 1 - \left[\frac{e^{-3} 3^0}{0!} + \frac{e^{-3} (3)^1}{1!} + \frac{e^{-3} (3)^2}{2!} + \frac{e^{-3} (3)^3}{3!} + \frac{e^{-3} (3)^4}{4!} + \frac{e^{-3} (3)^5}{5!} \right]$$

**MT-2005: Probability &
Statistics BSCS (Part B)**

(A, B, C, D, E, F, G, H, J, K)

Tuesday, 28th May, 2024**Course Instructors**Dr. Muhammad Usman Ashraf, Dr. Neelam,
Ms. Amara, Ms. Kiran

Serial No:

Final Exam

Max. Time: 80 min

Total Marks: 50

Signature of Invigilator

Student Name

Roll No.

Section

Signature

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3. Calculator sharing is strictly prohibited.
4. Area under the normal curve is attached for your information.

Question #1**[10 marks]**

Find variance and mode for the following data.

Daily wages	4-6	7-9	10-12	13-15	16-18	19-21
Number of employees	13	18	20	16	15	7

Question #2**[03+03+02=08 marks]**

- a) Six white and four black balls which are indistinguishable apart from colour are placed in a bag. If six balls are taken from the bag, find the probability that they are being three white and three black?
- b) In a group of 20 adults, 4 out of 7 women and 2 out of 13 men wear glasses. What is the probability that a person chosen at random from the group is a woman or someone who wear glasses?
- c) In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that the student did not take either of these subjects.

0-5
5-10
10-15
15-20
20-25

Question #3**[07 marks]**

The marks obtained out of 25 by 30 students of a class in the examination are given below:
20, 6, 23, 19, 9, 14, 15, 3, 1, 12, 10, 20, 13, 3, 17, 10, 11, 6, 21, 9, 6, 10, 9, 4, 5, 1, 5, 11, 7, 24.

Represent the above data as a grouped data taking the class interval 0-5, 5-10 and so on.

[2.5*4=10 marks]**Question #4**

Define following terms.

- Conditional Probability.
- Poisson distribution and Poisson process.
- Mutually exclusive and Exhaustive events.
- Cumulative distribution Function for Continuous Random Variable.

Question #5**[02+03=05 marks]**

Consider the density function

$$f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

- Evaluate k.
- Find $F(x)$ and use it to evaluate $P(0.3 < X < 0.6)$

Question #6**[06+04=10 marks]**

The average grade for an exam is 74, and the standard deviation is 7. If 12% of the class is given A's, and the grades are curved to follow a normal distribution, what is the lowest possible A and the highest possible B? Also find the sixth decile and 90th percentile.

Q1 (10 marks)

Just mode = 11
zero

1 mark

Daily wages	f	X	fx	fx ²	(x - $\bar{x}$) ²	f(x - $\bar{x}$) ²
4-6	13	5	65	325	45.9	596.7
7-9	18	8	144	1152	14.25	256.5
10-12	20	11	220	2420	0.601	12.02
13-15	16	14	224	3136	4.95	79.21
16-18	15	17	255	4335	27.3	409.5
19-21	7	20	140	2800	67.65	473.55
	89		1048	14168		1827.48

formula: 1 mark

5 marks
$$\text{Var} = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2$$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{1048}{89} - \left(\frac{1048}{89} \right)^2$$

$$= \frac{1048}{89} - \frac{138.65}{89} = 11.54$$

2 mark

if answer is incorrect then 1.5

5 marks
$$\text{Mode} = l + \frac{(f_m - f_1)}{(f_m - f_1) + (f_m - f_2)} \times h$$

$$f_m \rightarrow 1 \text{ mark} = 9.5 + \frac{(20 - 18)}{(20 - 18) + (20 - 16)} \times 3$$

$$f_1 \rightarrow 1 \text{ mark}$$

$$f_2 \rightarrow 1 \text{ mark}$$

$$= 9.5 + \frac{2}{2 + 4} \times 3$$

$$= 9.5 + \frac{6}{6} = 10.5 \rightarrow 1 \text{ mark}$$

ignore $10 + 1 = 11$

$$\frac{\sum f(x - \bar{x})^2}{\sum f}$$

$$= \frac{1827.48}{89}$$

$$= 20.54$$

missing

$$10 + 1 = 11$$

$$= l + \frac{(f_m - f_1) \times h}{2f_m - f_1 - f_2}$$

$$= 9.5 + \frac{2 \times 3}{40 - 18 - 16}$$

$$= 9.5 + \frac{6}{6}$$

$$= 9.5 + 1 = 10.5$$

Q2 2 of 6

(3 marks)

(a) $6W + 4B = 10$

$\binom{10}{C \quad 6} = 210$ $n=6$

Let $E = 3$ white and 3 black balls

$n(E) = \binom{6}{C \quad 3} \cdot \binom{4}{3} = 20 \times 4 = 80$

$P(E) = \frac{80}{210} = \frac{8}{21} = 0.38095$ → 1 mark

(3 marks)

(b) 20 adults, $\frac{7}{13}$ women and $\frac{2}{13}$ men wear glasses.

Prob = a person chosen at random from group is a woman or someone who wear glasses

$E_1 = \{ \text{person chosen is a woman} \} \Rightarrow P(E_1) = \frac{7}{20}$

$E_2 = \{ \text{person chosen who wear glasses} \} \Rightarrow P(E_2) = \frac{6}{20}$

$0.35 + 0.42 - 0.57 \rightarrow \text{zero}$

$P(\text{The women who wear glasses}) = \frac{4}{20}$

$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$
 $\frac{7}{20} + \frac{6}{20} - \frac{4}{20} = \frac{9}{20} = 0.45$ → 2 marks

(2 marks) 0.35, 0.3, 0.2 → 1 mark

(c) 100 students
 $54M + 69 \text{ Hitz} + 35 \text{ Hist} + M$

If 1 student is selected at random, find prob student didn't take either of these subjects

Prob that either of these subjects = $P(M \cup H) = \frac{54+69-35}{100}$

Prob that didn't " " " = $1 - P(M \cup H)$
 $= P(M' \cap H')$ → 1 mark

$= 1 - \frac{88}{100} = \frac{12}{100} = \frac{3}{25} = 0.12$ → 1 mark

83 → 07 marks

20, 6, 23, 19, 9, 14, 15, 3, 1, 12, 10, 20, 13, 3, 17, 10, 11, 6, 21, 9, 6, 10, 9, 4, 5, 5, 1, 5, 11, 7, 24

not included

C.I	Tally	f	(marks)
5-10		5	1
10-15		4	1
15-20		3	1
20-25		5	1
	ignore	30	

marks
9.0

7 marks

Q4 Conditional Prob : (a) $P(A|B) = \frac{P(A \cap B)}{P(B)}$ 2 marks

(b) Poisson dist : $e^{-\mu} \frac{\mu^x}{x!}$, Poisson Process : $e^{-\lambda t} \frac{(\lambda t)^x}{x!}$

(c) $P(A_i \cap A_j) = \emptyset$, $P(A_1 \cup A_2 \dots \cup A_k) = S$
 $i \neq j$

(d) $F(x) = \int_{-\infty}^x f(x) dx$; $F(\infty) = 0$
 $F(-\infty) = 1$

Q5

$$f(x) = \begin{cases} k\sqrt{x} & ; 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\int_0^1 k x^{1/2} dx = k \left[\frac{x^{3/2}}{3/2} \right]_0^1$$
$$= \frac{2}{3} k [1 - 0] = 1,$$

$$k = 3/2. \quad (2)$$

$$f(x) = \frac{3}{2} x^{1/2} \quad ; 0 < x < 1$$

$$F(x) = \int_0^x \frac{3}{2} x^{1/2} dx = \frac{3}{2} \left[\frac{x^{3/2}}{3/2} \right]_0^x$$

$$F(x) = x^{3/2} \quad ; 0 < x < 1$$

(1) $= 1$ ~~elsewhere~~
 $x > 1$

$$P(0.3 < x < 0.6) = F(0.6) - F(0.3)$$

$$= (0.6)^{3/2} - (0.3)^{3/2}$$

(2)

$$= 0.4648 - 0.1643$$

$$= 0.3005$$

$$\mu = 74$$

$$\sigma = 7$$

12% of class is given A. 6+2+2

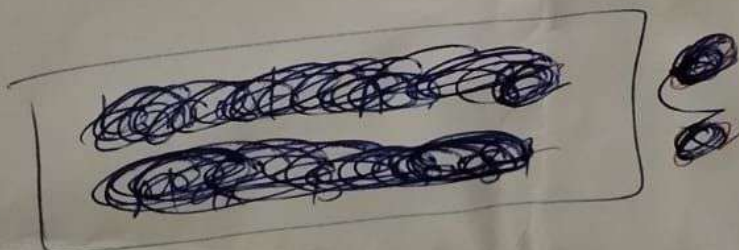
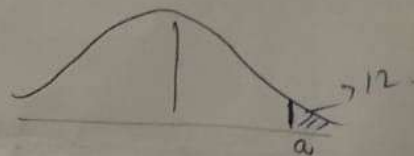
$$P(X \geq a) = 0.88$$

$$P\left(Z \leq \frac{a-74}{7}\right) = 0.88$$

$$P(Z \leq 1.18) = 0.88$$

$$1.18 = \frac{a-74}{7}$$

$$a = 74 + 7(1.18) = 82.26$$



$$P(X \geq b) = 0.60$$

$$P(Z \leq 0.25) = 0.60$$

$$0.25 = \frac{a-74}{7}$$

$$D_6 = a = 75.75$$

$$P_{90} = P(Z \leq 1.28) = 0.90$$

$$1.28 = \frac{a-74}{7}$$

$$P_{90} = 82.96$$